



Ofqual QAN	600/3055/0
Quartz ID	1158
Qualification	Level 5 Diploma in Computing

Unit ID	2182
Assessment	Mobile App Development
Assessment Type	Global Assignment
Cycle	Spring 2026

Assignment title: Building a Task Management App

Important notes

- Please refer to the *Assignment Presentation Requirements* for advice on how to set out your assignment. These can be found on the NCC Education website. Hover over 'About Us' on the main menu and then navigate to 'Policies and Procedures' then scroll to the 'Student Support' area.
- You **must** read the NCC Education document *Academic Misconduct Policy* and ensure that you acknowledge all the sources that you use in your work. These documents are available on the NCC Education website. Hover over 'About Us' on the main menu and then navigate to 'Policies and Procedures' then scroll to the 'Student Support' area.
- You **must** complete the *Statement and Confirmation of Own Work*. The form is available on the NCC Education website. Hover over 'About Us' on the main menu and then navigate to 'Policies and Procedures' then scroll to the 'Student Support' area.
- **Please make a note of the recommended word count as stated in the brief. The maximum word count limit is the recommended work count plus 10% leeway. Submissions which exceed the maximum word count limit will be marked only up to the maximum word count limit. Any text that exceeds the maximum word count limit will not be marked. You must follow the note on word count as stated in the Submission Requirements section of the brief.**
- You must submit a paper copy and digital copy (on disk or similarly acceptable medium). Media containing viruses, or media that cannot be run directly, will result in a fail grade being awarded for this assessment.
- All electronic media will be checked for plagiarism.

Scenario

Project Overview:

You have been assigned to create a fitness tracking mobile application for Android devices. The app should enable users to track their daily exercise routines, log workouts, set fitness goals and monitor progress over time.

To encourage your creativity and cater to different levels of proficiency, you should have some choice over the types of input you can provide for the app. The basic functionality should include input options for activities such as running, cycling, and weightlifting, with the app providing detailed statistics and analytics for each activity. Additionally, to achieve higher grades, you could extend the functionality by incorporating other activities of your choice and developing more advanced analytical features or personalised insights based on the user's activity data. This approach allows you to work within a foundational framework while encouraging you to innovate and customise your project.

The app also requires allowing users to register and log in via the app. All user data will be securely stored on a MySQL Server Database via a PHP-based web service.

The application should prioritise user experience with a visually appealing interface and seamless navigation.

Task 1: Understanding the Scope and Position of Mobile Apps – 15 Marks

- a) *Key Characteristics of Mobile Apps*: Discuss the essential features and functionalities commonly found in mobile applications, using examples from the scenario app where users input activities like running, cycling, or weightlifting. For instance, describe how offline accessibility allows users to log their workouts without an internet connection and how the app syncs data once connectivity is restored. Explain how push notifications can remind users to log their activities or notify them of their progress and achievements. Highlight the importance of touch-based interfaces for easy navigation, allowing users to quickly input their activity details and view their statistics and analytics.
- b) *Scope and Limitations*: Evaluate the potential scope of mobile apps in the fitness tracking industry and discuss the limitations, including device compatibility, screen sizes, and hardware constraints. For example, in the context of a fitness tracking mobile application, explore how such apps can monitor activities like running, cycling, and weightlifting, providing users with detailed statistics and analytics. Discuss limitations such as adapting to various screen sizes for an optimal user experience, and addressing hardware constraints like battery life and sensor accuracy that could impact the app's performance and reliability.
- c) *Developing for Multiple Platforms*: Discuss the challenges and strategies for developing apps that are compatible with multiple platforms, such as iOS and Android.

Task 2: Understanding Development Architectures, Platforms, Languages and Tools – 15 Marks

- a) *Key Characteristics of Native Apps*: Explain the advantages and disadvantages of developing native apps, using a fitness tracking mobile application as the scenario. For example, in the context of a fitness tracking app that monitors activities like running, cycling, and weightlifting, discuss the advantages of native apps such as performance optimisation, which allows for smoother and faster tracking of real-time data, and access to platform-specific features like GPS, accelerometers, and health data integration. Additionally, highlight the disadvantages, such as the need for separate development efforts for iOS and Android platforms, which can increase costs and development time. Discuss how these factors can impact the overall user experience and functionality of a fitness tracking app.
- b) *Key Characteristics of Cross-Platform Development Frameworks*: Compare popular cross-platform development frameworks like React Native and Xamarin, highlighting their features and limitations in the context of developing a fitness tracking mobile application. Consider factors such as performance, compatibility with fitness tracking sensors and APIs, user interface customisation options, and ease of integrating platform-specific features such as geolocation and push notifications.
- c) *Current Trends in Development Tools*: Summarise the latest trends in mobile app development tools, including integrated development environments (IDEs), version control systems and automated testing frameworks.
- d) *Native vs. Cross-Platform Approaches*: Evaluate the relative merits of native and cross-platform development approaches for developing the Fitness Tracking App, considering factors like time-to-market, performance and maintainability.

Task 3: Planning and Designing the Mobile Application – 15 Marks

- a) *SDLC and Mobile Apps*: Explain how the Software Development Life Cycle (SDLC) applies to mobile app development, including stages like requirements analysis, design, implementation, testing and deployment.
- b) *Specification and Design*: Employ appropriate methods to specify and design the Fitness Tracking App, considering user requirements, user interface design principles, and usability guidelines. Provide clear explanations for design decisions and justify choices made throughout the specification and design process. The specification should outline the basic structure such input data requirements, system architecture, user interface wireframes, and interaction flows.

Task 4: Developing Application Functionality – 40 Marks

- a) Using Android Studio: Install and utilise Android Studio as the primary IDE for developing the Fitness Tracking App, including setting up the project, creating layouts and writing code.
- b) User Interface Elements: Implement common user interface elements like RecyclerView, FloatingActionButton and Geolocation components to construct the application interface.
- c) Database: Create a Database using MySQL to store and retrieve application data such as user data, exercise routines etc.

Task 5: Employing Object-Oriented Techniques – 15 Marks

- a) Object Orientation in Kotlin and Android: Discuss how Kotlin and Android utilise Object-Oriented Programming (OOP) principles like encapsulation, inheritance and polymorphism to enhance code quality and development productivity.
- b) Inheritance Usage: Make appropriate use of inheritance in designing and coding the Fitness Tracking App, demonstrating how it facilitates code reuse and maintainability. Explain THREE (3) examples of how / where you have employed inheritance in your app.

Submission:

- The completed Fitness Tracking App project along with its source code
- A written report addressing the assessment criteria outlined above

Submission requirements

- Your submitted work must be word-processed, justified, and double-spaced.
- You will need to hand in a printed copy of your work, with the *Statement and Confirmation of Own Work* form attached.
- You will need to submit an electronic copy.
- You will need to include your name and NCC Education student number in the header on every page of your work.
- Diagrams and materials associated with the tasks above should be presented in a word-processed document.
- Ensure that all references and citations adhere to the Harvard Style referencing format. This includes accurately citing sources within the text and compiling a complete reference list at the end of the document according to the Harvard Style guidelines. Consistency and accuracy in referencing are essential to maintain academic integrity and provide proper credit to the sources used in the assignment.

Word Count

The word count for this assignment is **2,500 words**.

You **MUST** state the word count in the header of your assignment and declare the number of words on your assignment cover sheet. You **MUST** confirm that you have not exceeded the stipulated word limit by more than 10%.

Any words written beyond the stipulated limit will not be read by the examiners; no marks will be awarded, and no feedback provided for any text beyond the prescribed limit.

In accordance with accepted academic practice, the notion of word count includes the following without exception:

- All titles or headings that form part of the actual text. This does not include the fly (or cover) page or reference list.
- All words that form the actual work.
- All words forming the titles for figures, tables, and boxes. However, this does not include the text within boxes, tables, or figures themselves.
- All in-text (bracketed) references.
- All directly quoted material.

Candidate checklist

Please use the following checklist to ensure that your work is ready for submission.

Have you read the NCC Education document *Academic Misconduct Policy* and ensured that you have acknowledged all the sources that you have used in your work? ☐

Have you completed the *Statement and Confirmation of Own Work* form and attached it to your assignment? **You must do this.** ☐

Have you ensured that your work has not gone over or under the recommended word count by more than 10%? ☐

Have you ensured that your work does not contain viruses and can be run directly? ☐